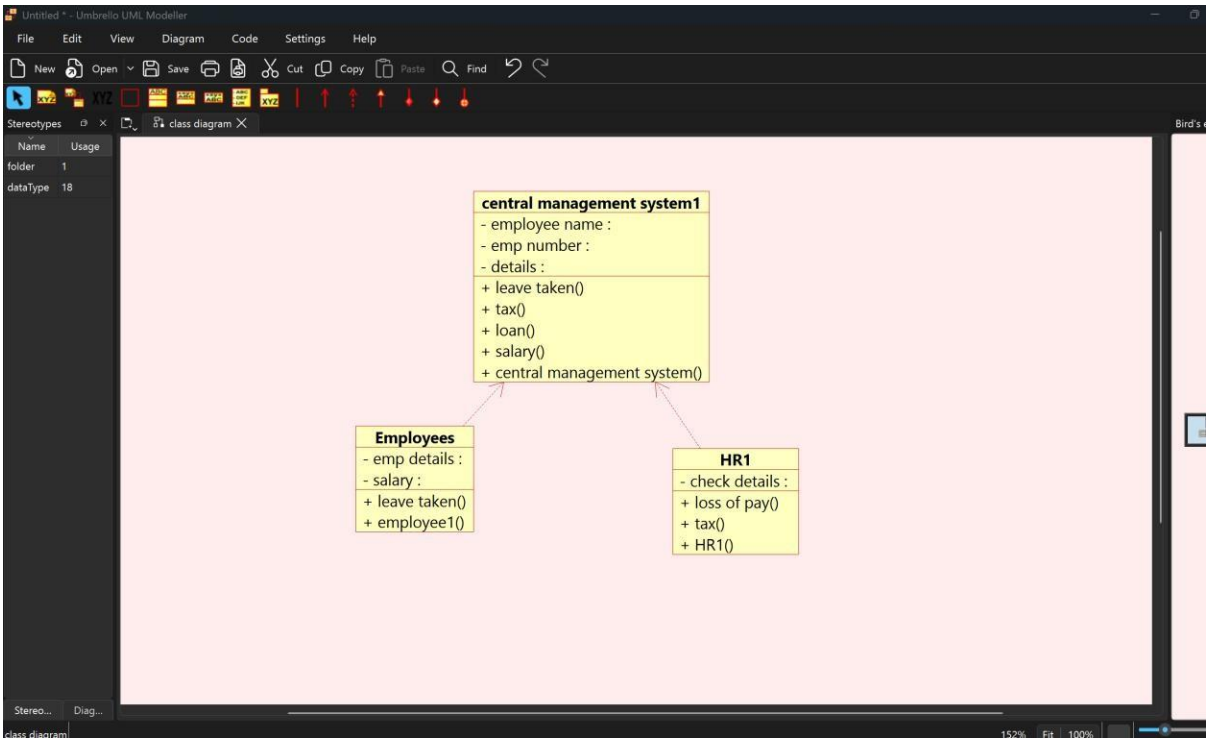
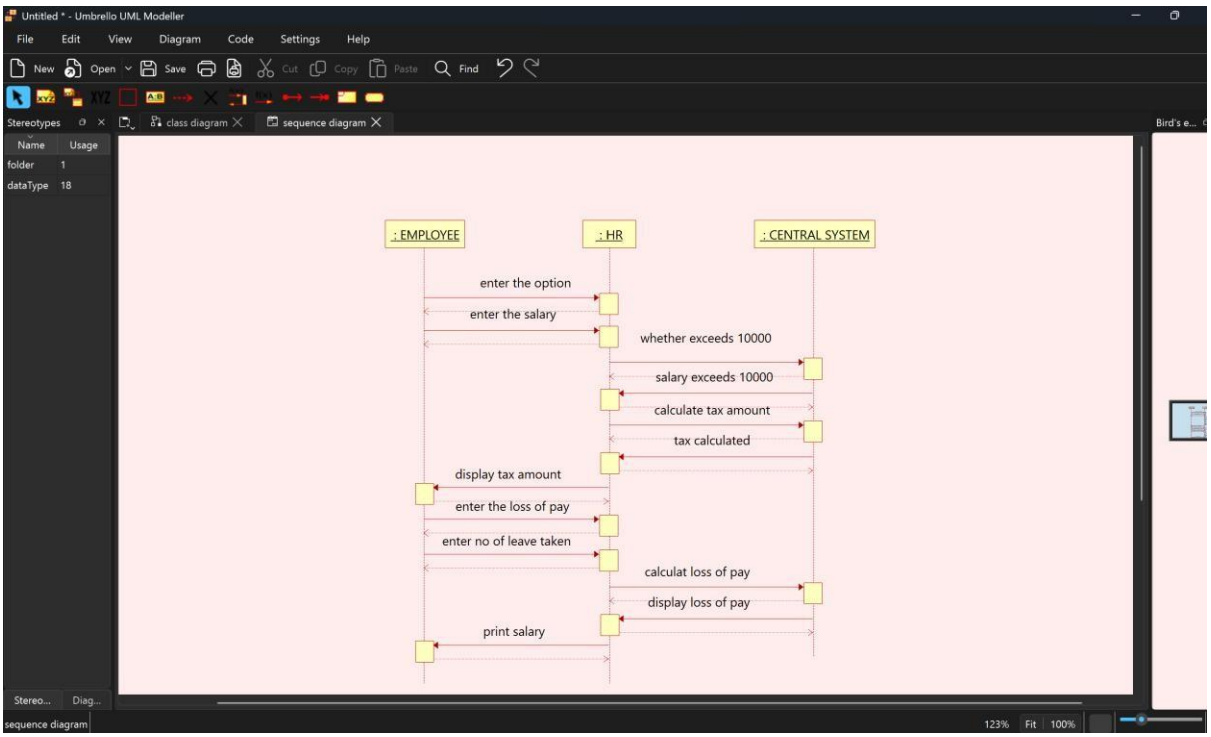
Ex No: 8 – SOFTWARE PERSONNEL MANAGEMENT SYSTEM

NAME: Arava Raja (192365033)

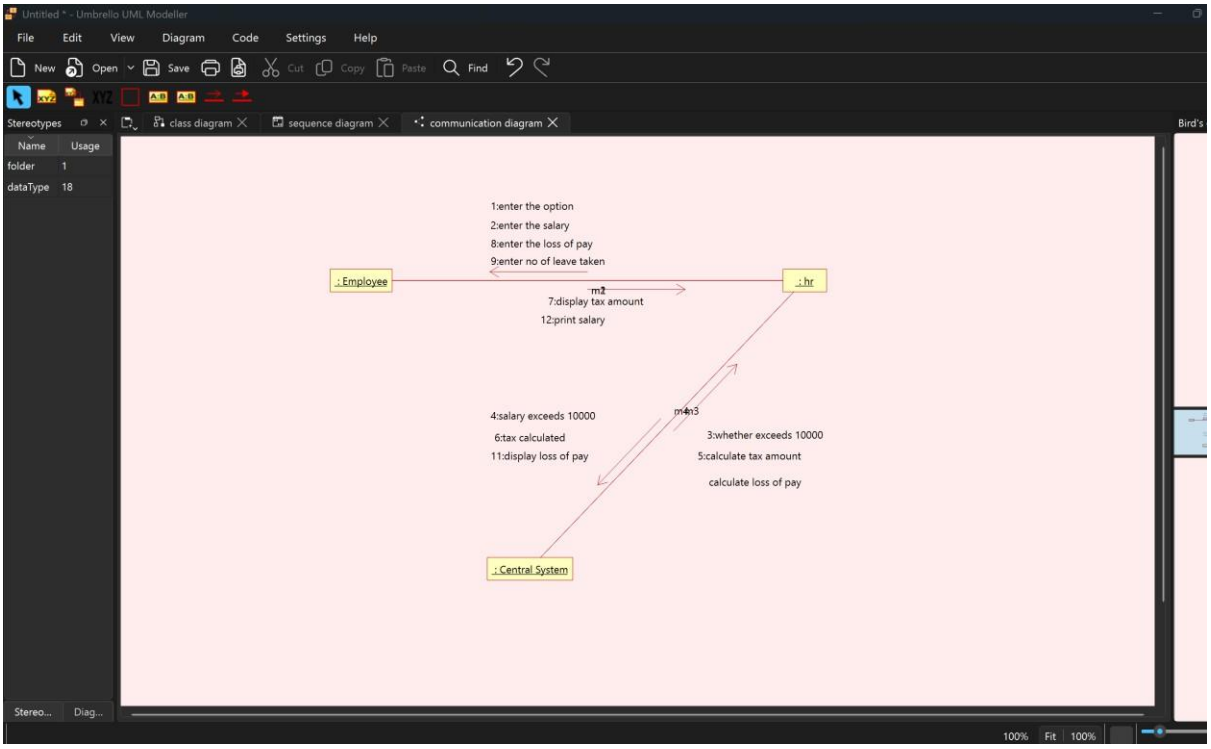
CLASS DIAGRAM:



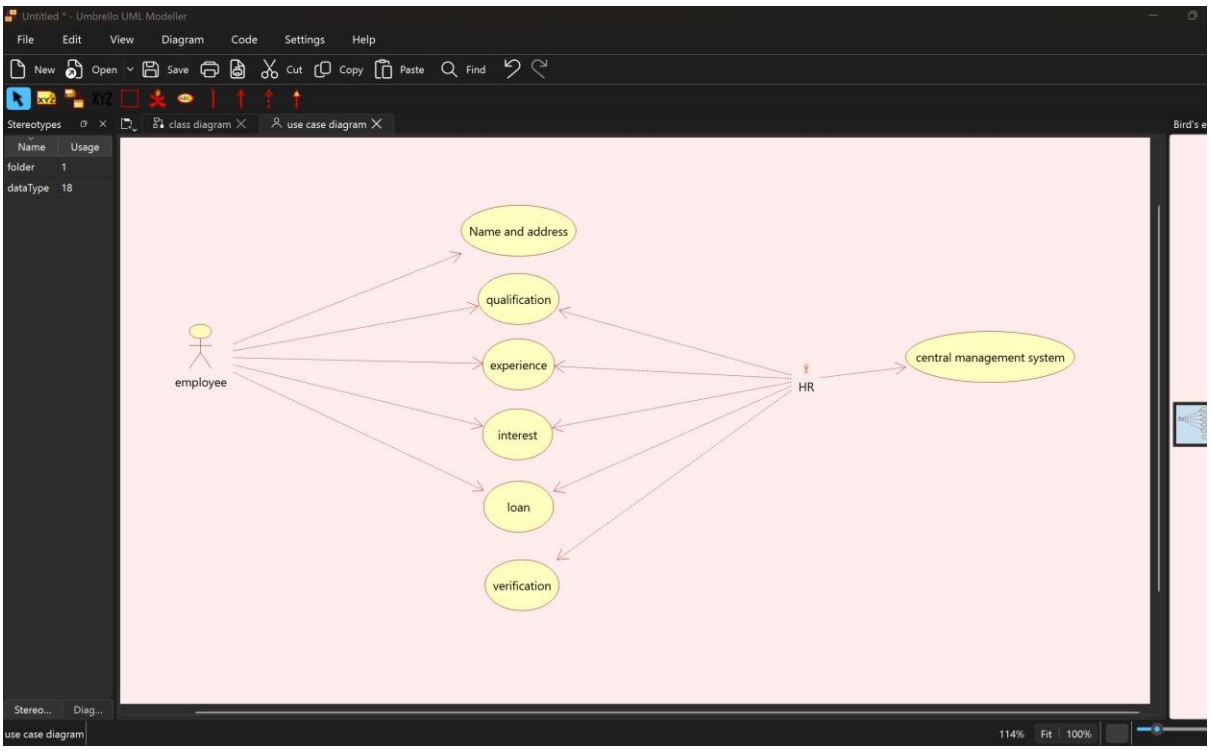
SEQUENCE DIAGRAM:



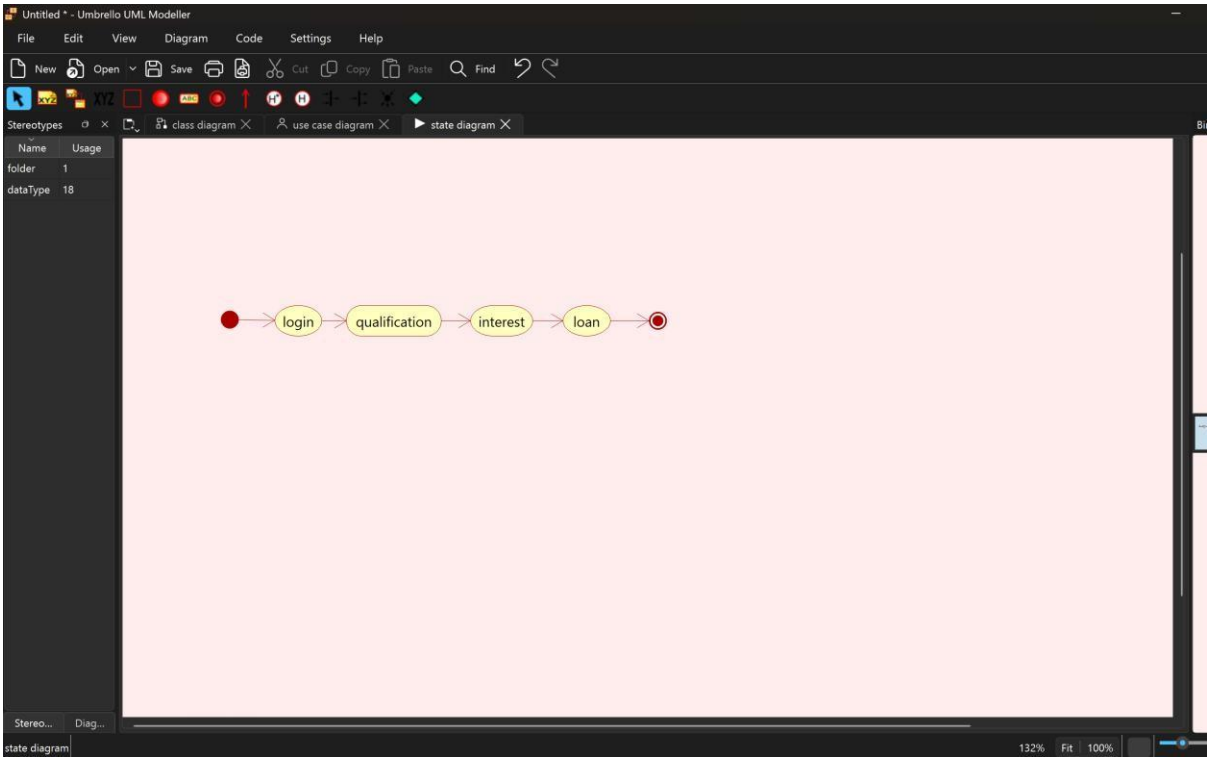
COLLABORATION DIAGRAM:



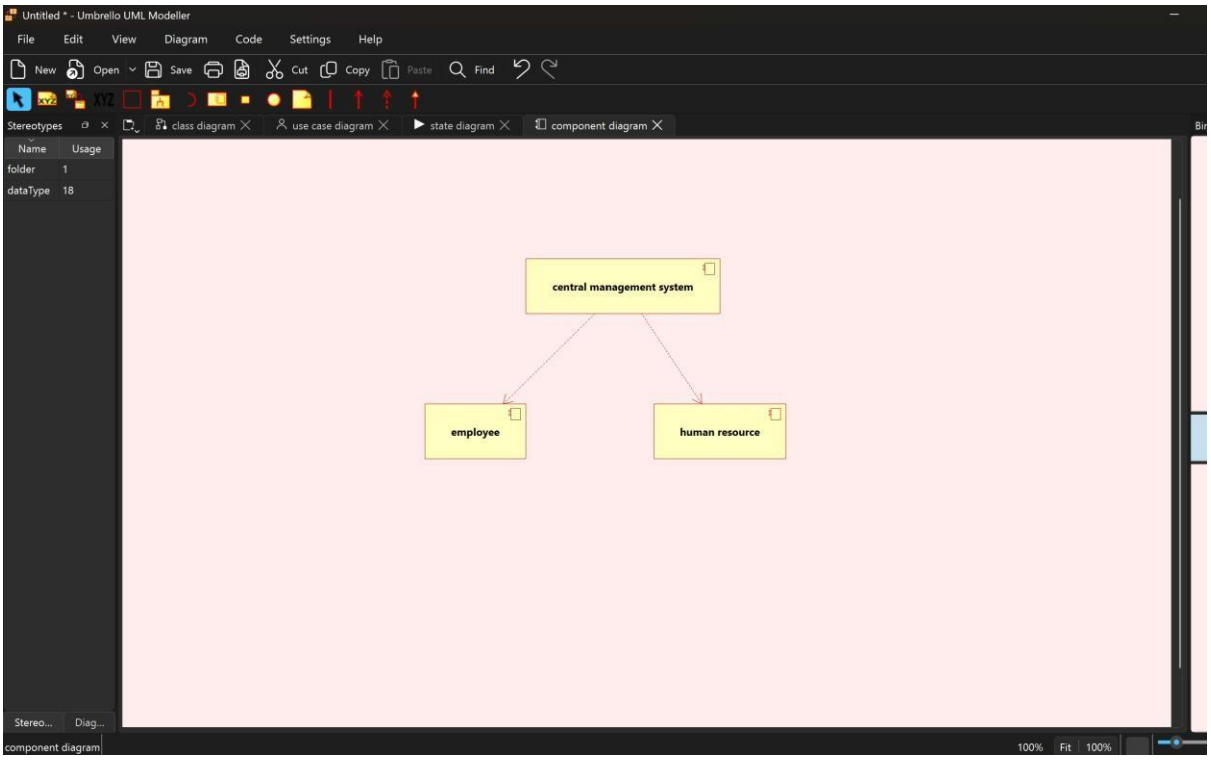
USE CASE DIAGRAM:



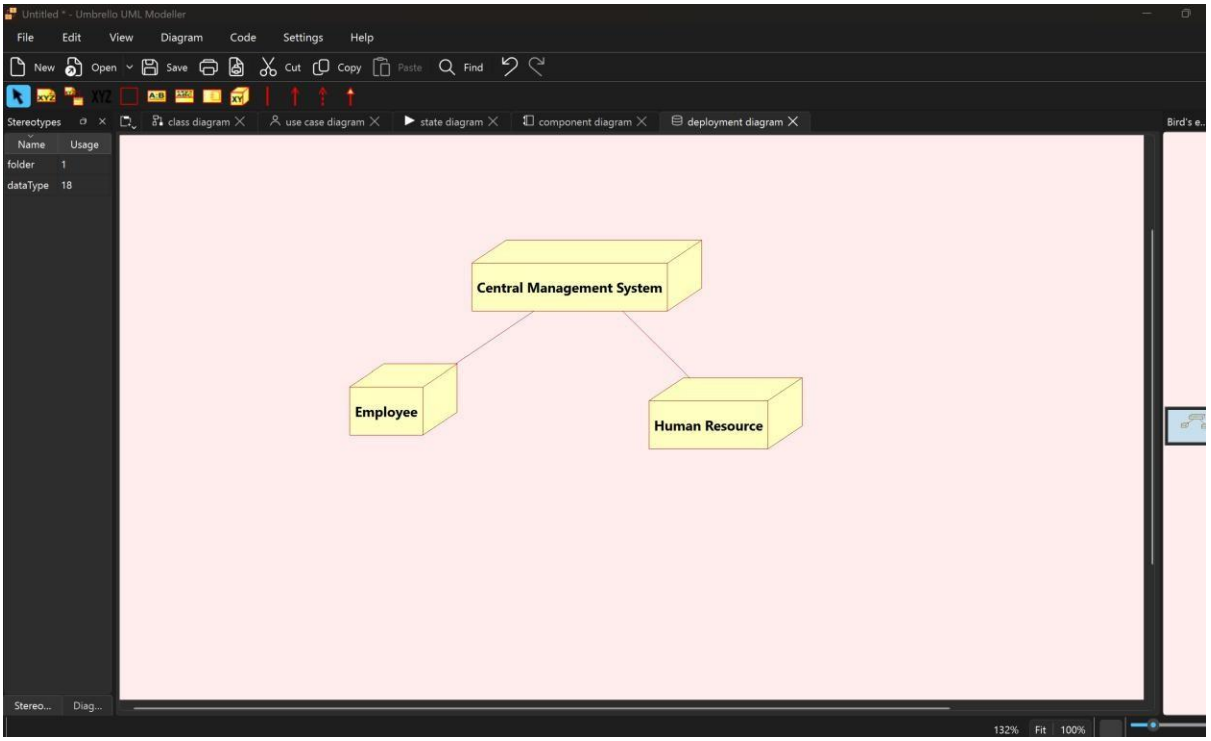
STATE DIAGRAM:



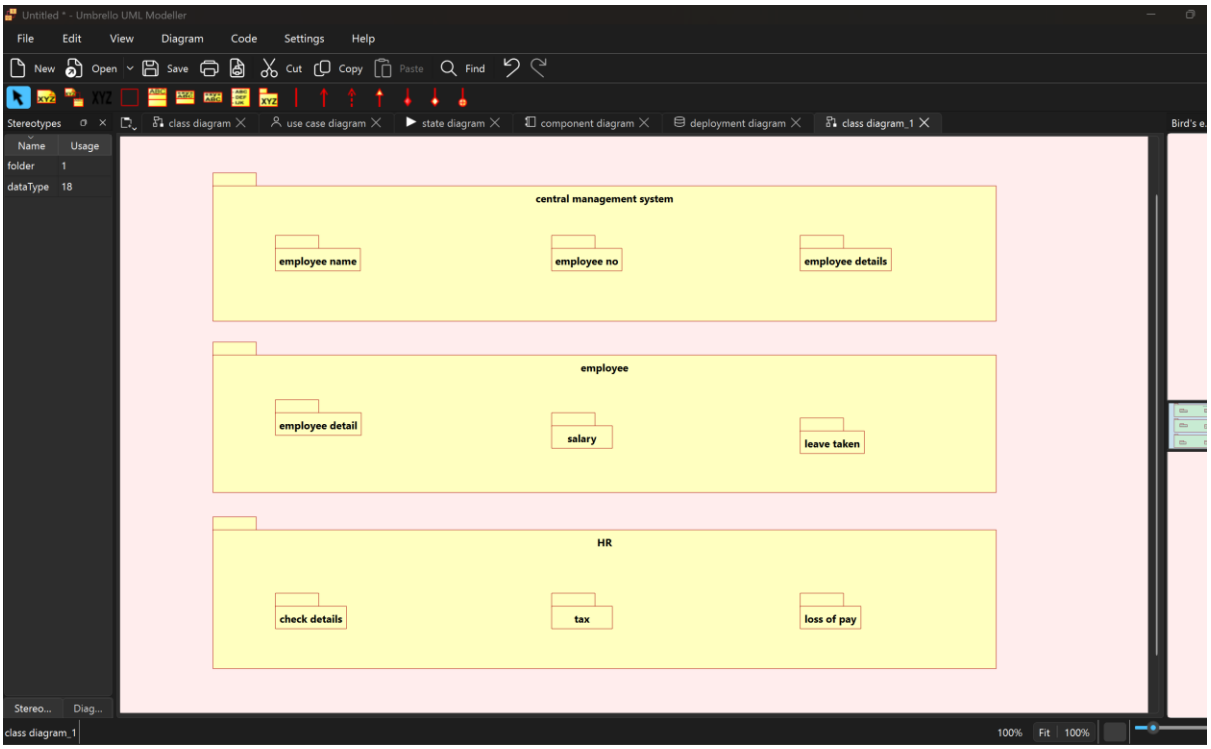
COMPONENT DIAGRAM:



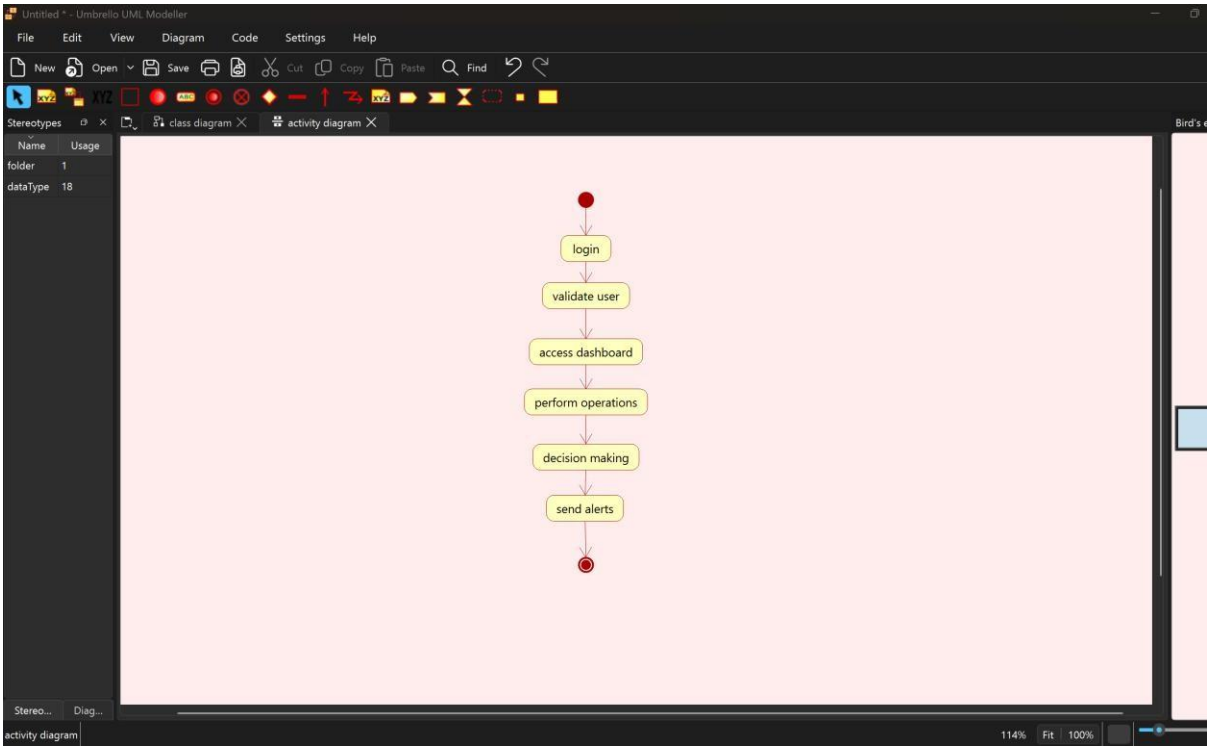
DEPLOYMENT DIAGRAM:



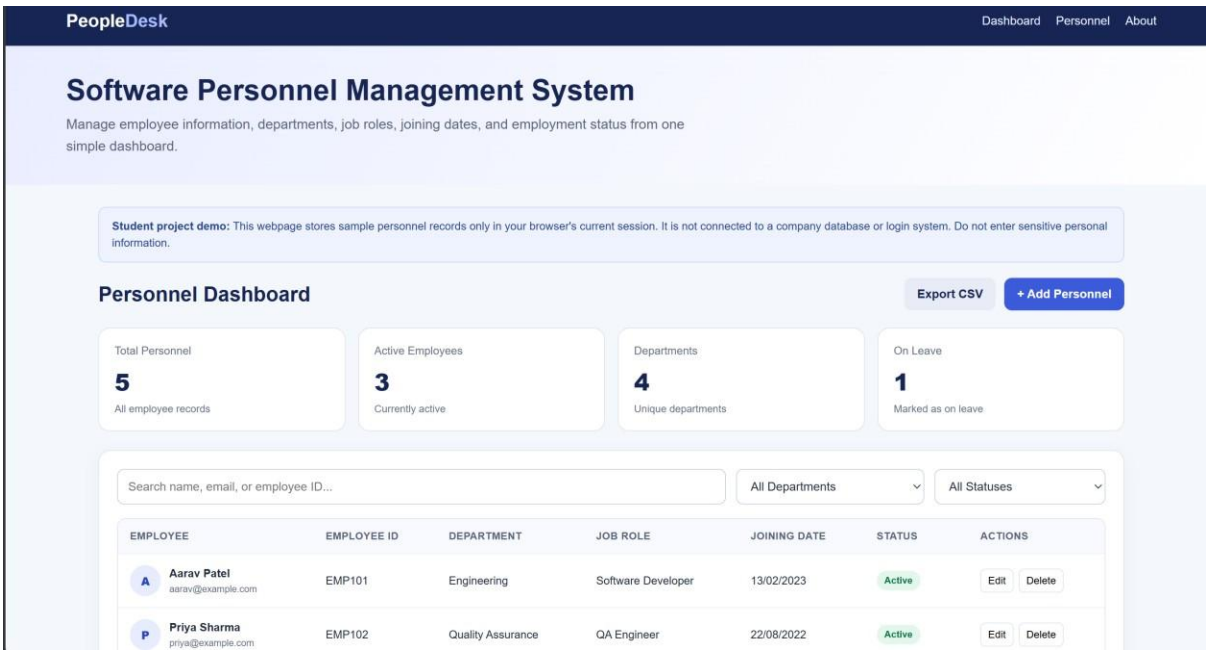
PACKAGE DIAGRAM

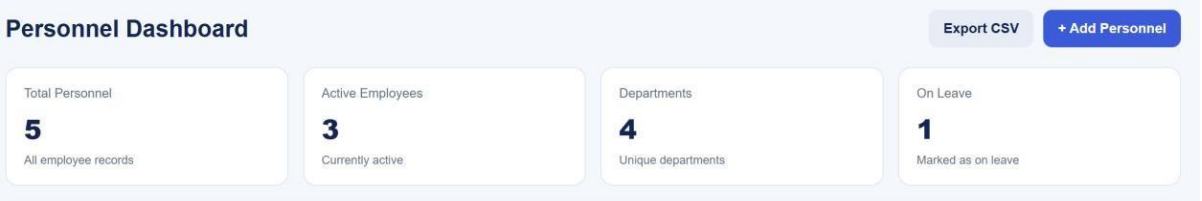


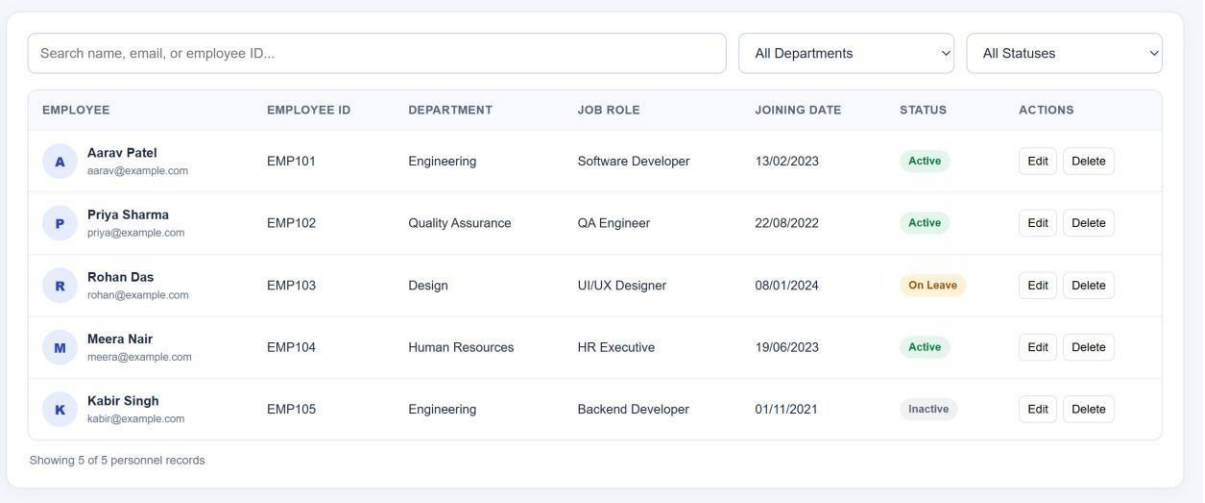
ACTIVITY DIAGRAM:

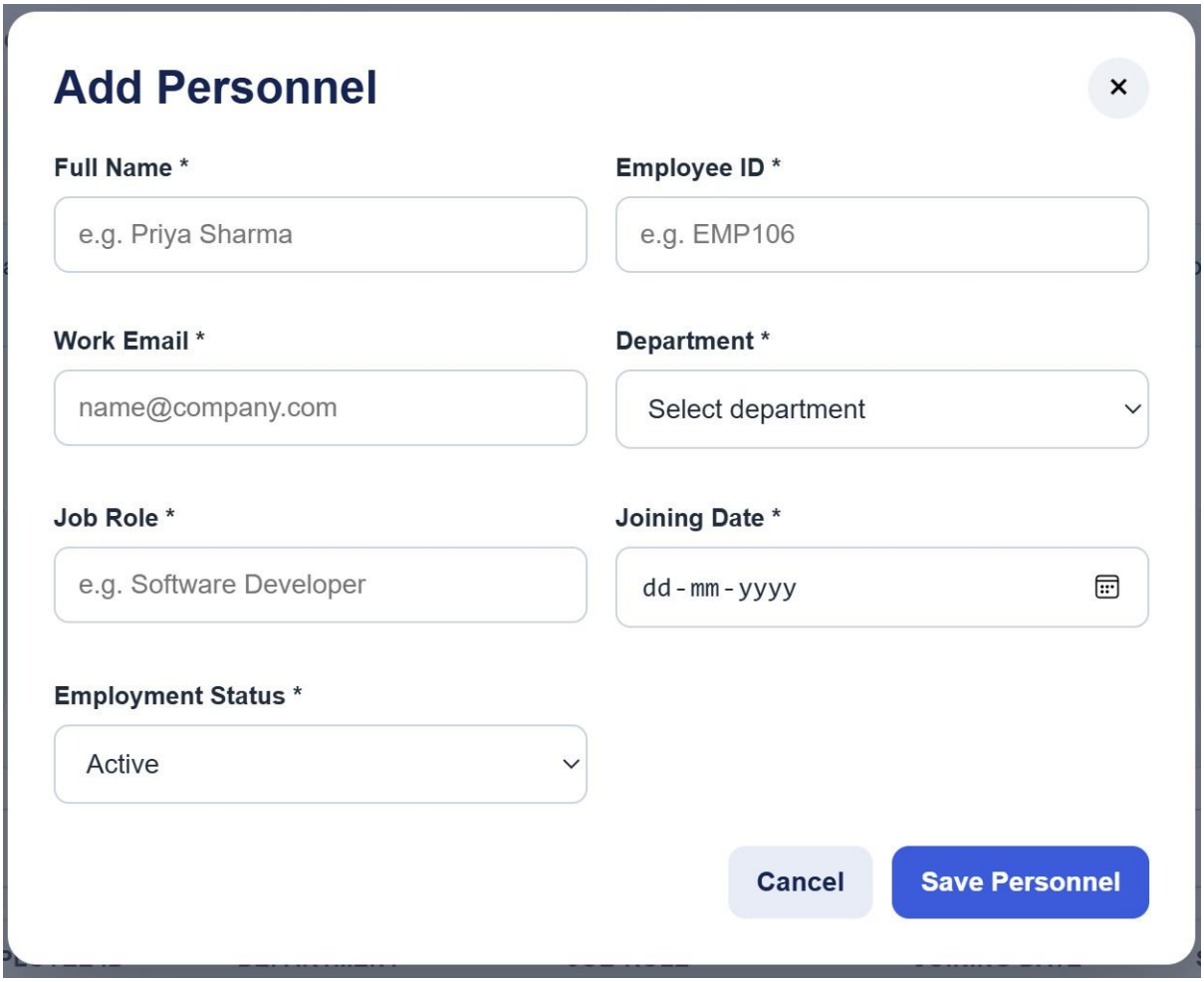


WEBPAGE FOR SOFTWARE PERSONNEL MANAGEMENT SYSTEM

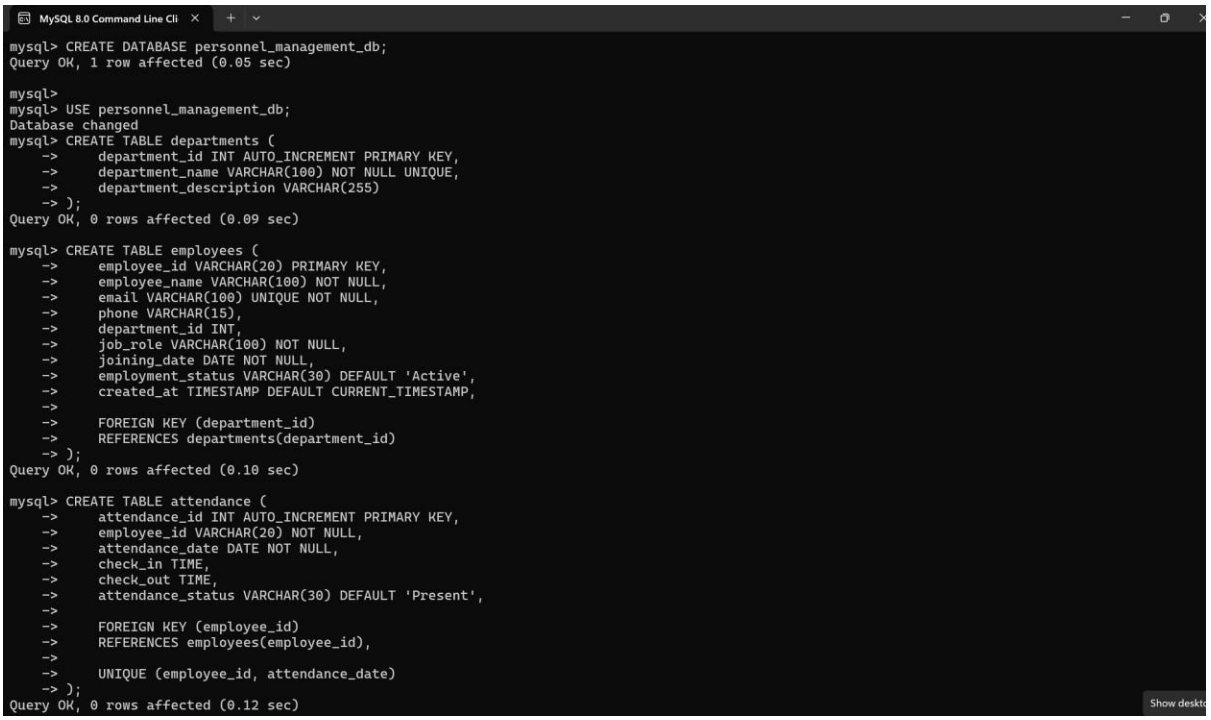



Personnel Dashboard
Export CSV
+ Add Personnel
Total Personnel
5
All employee records
Active Employees
3
Currently active
Departments
4
Unique departments
On Leave
1
Marked as on leave


Search name, email, or employee ID...
All Departments
All Statuses
EMPLOYEE
EMPLOYEE ID
DEPARTMENT
JOB ROLE
JOINING DATE
STATUS
ACTIONS
A
Aarav Patel
aarav@example.com
EMP101
Engineering
Software Developer
13/02/2023
Active
Edit
Delete
P
Priya Sharma
priya@example.com
EMP102
Quality Assurance
QA Engineer
22/08/2022
Active
Edit
Delete
R
Rohan Das
rohan@example.com
EMP103
Design
UI/UX Designer
08/01/2024
On Leave
Edit
Delete
M
Meera Nair
meera@example.com
EMP104
Human Resources
HR Executive
19/06/2023
Active
Edit
Delete
K
Kabir Singh
kabir@example.com
EMP105
Engineering
Backend Developer
01/11/2021
Inactive
Edit
Delete
Showing 5 of 5 personnel records


Add Personnel
Full Name *
e.g. Priya Sharma
Employee ID *
e.g. EMP106
Work Email *
name@company.com
Department *
Select department
Job Role *
e.g. Software Developer
Joining Date *
dd-mm-yyyy
Employment Status *
Active
Cancel
Save Personnel

MYSQL DATABASE:



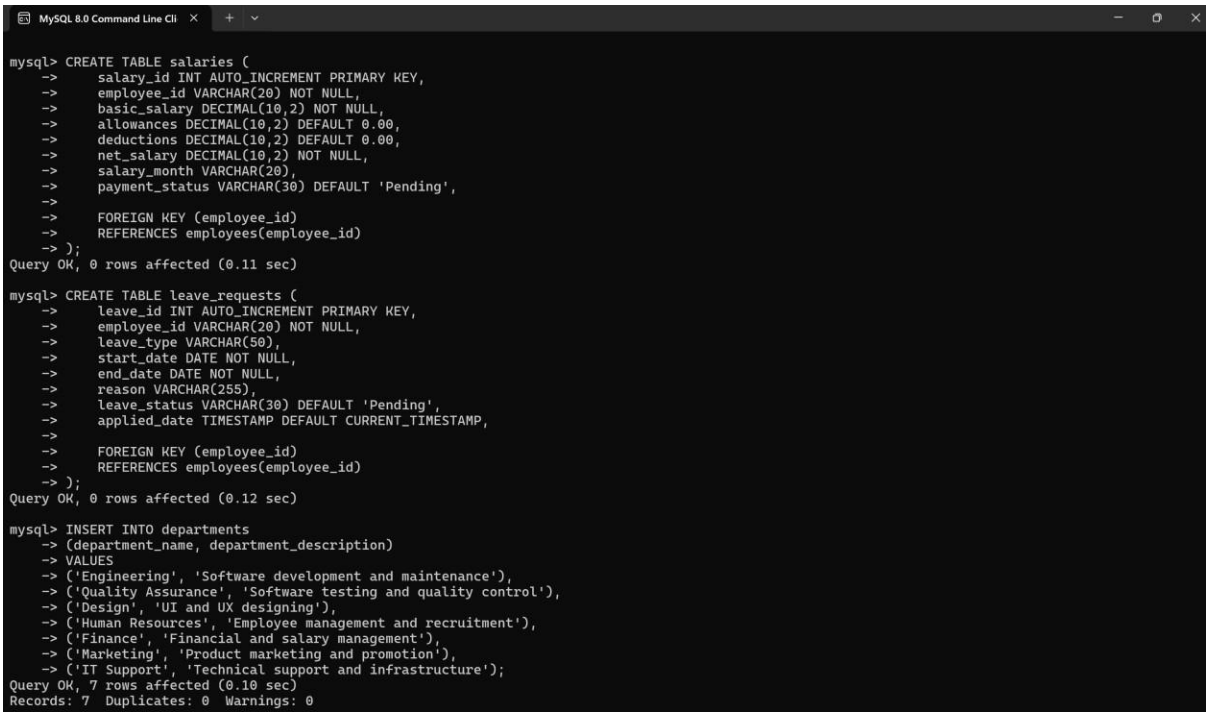
```
MySQL 8.0 Command Line Cli

mysql> CREATE DATABASE personnel_management_db;
Query OK, 1 row affected (0.05 sec)

mysql>
mysql> USE personnel_management_db;
Database changed
mysql> CREATE TABLE departments (
    ->     department_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     department_name VARCHAR(100) NOT NULL UNIQUE,
    ->     department_description VARCHAR(255)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE employees (
    ->     employee_id VARCHAR(20) PRIMARY KEY,
    ->     employee_name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL,
    ->     phone VARCHAR(15),
    ->     department_id INT,
    ->     job_role VARCHAR(100) NOT NULL,
    ->     joining_date DATE NOT NULL,
    ->     employment_status VARCHAR(30) DEFAULT 'Active',
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->
    ->     FOREIGN KEY (department_id)
    ->     REFERENCES departments(department_id)
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> CREATE TABLE attendance (
    ->     attendance_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     employee_id VARCHAR(20) NOT NULL,
    ->     attendance_date DATE NOT NULL,
    ->     check_in TIME,
    ->     check_out TIME,
    ->     attendance_status VARCHAR(30) DEFAULT 'Present',
    ->
    ->     FOREIGN KEY (employee_id)
    ->     REFERENCES employees(employee_id),
    ->
    ->     UNIQUE (employee_id, attendance_date)
    -> );
Query OK, 0 rows affected (0.12 sec)
```

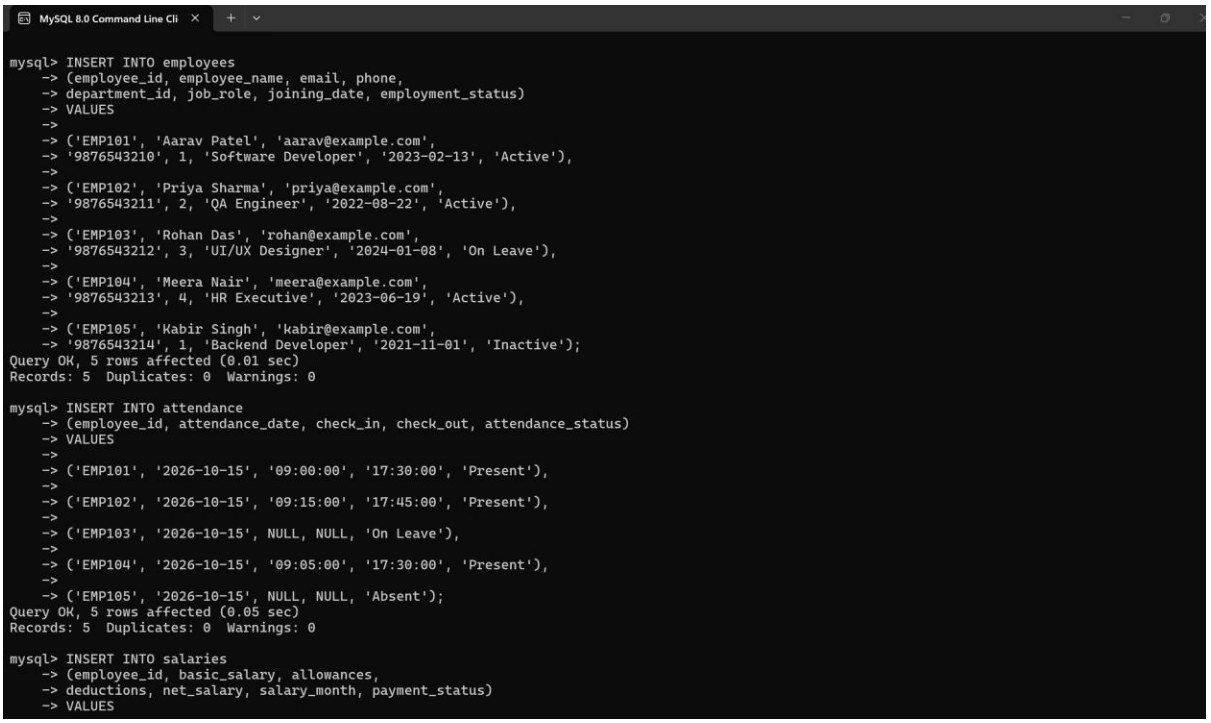


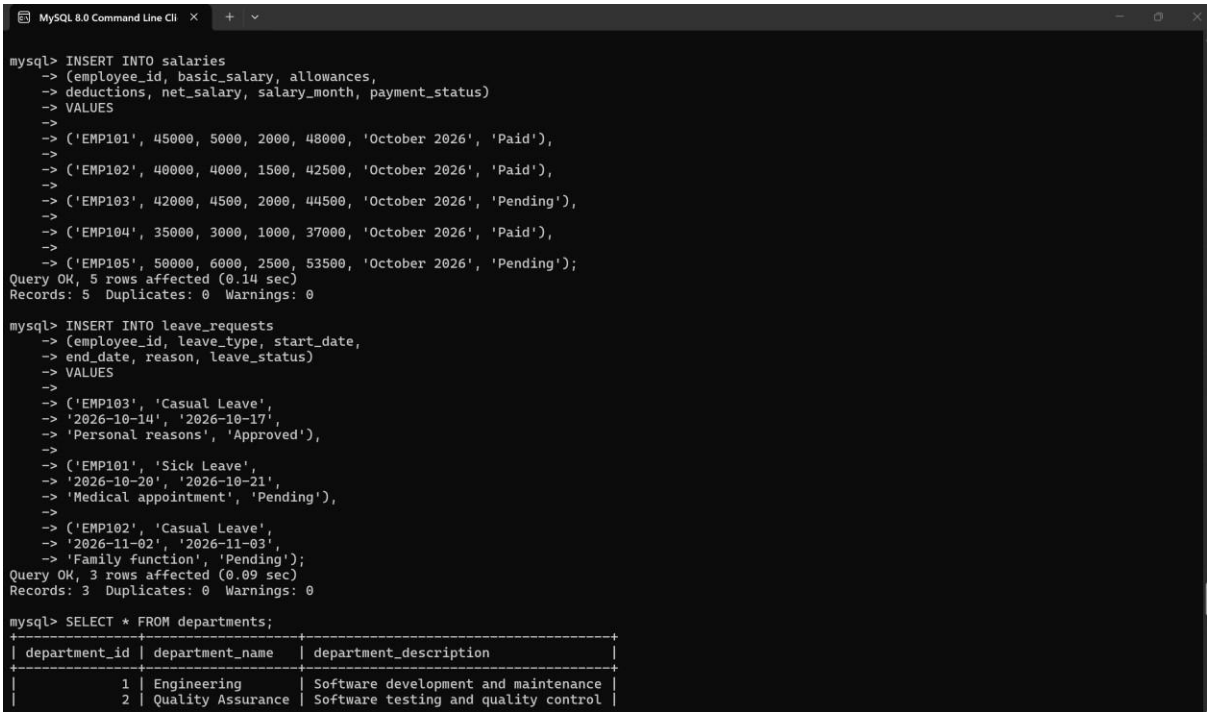
```
MySQL 8.0 Command Line Cli

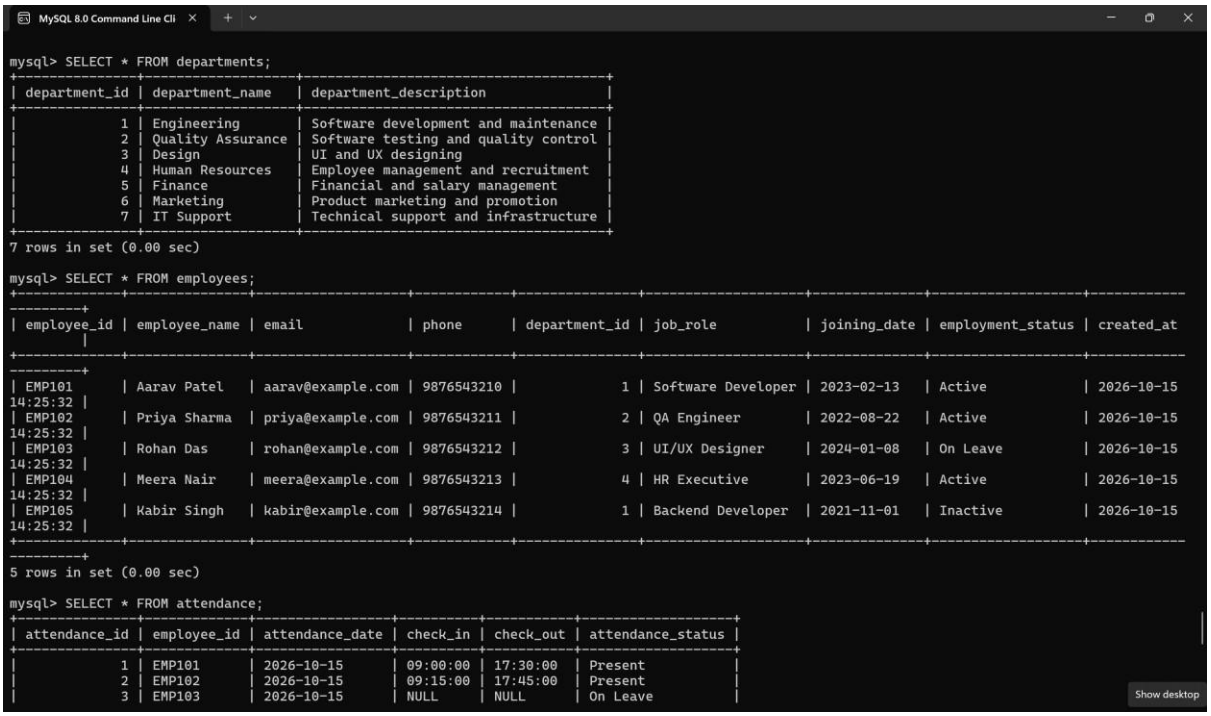
mysql> CREATE TABLE salaries (
    ->     salary_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     employee_id VARCHAR(20) NOT NULL,
    ->     basic_salary DECIMAL(10,2) NOT NULL,
    ->     allowances DECIMAL(10,2) DEFAULT 0.00,
    ->     deductions DECIMAL(10,2) DEFAULT 0.00,
    ->     net_salary DECIMAL(10,2) NOT NULL,
    ->     salary_month VARCHAR(20),
    ->     payment_status VARCHAR(30) DEFAULT 'Pending',
    ->
    ->     FOREIGN KEY (employee_id)
    ->     REFERENCES employees(employee_id)
    -> );
Query OK, 0 rows affected (0.11 sec)

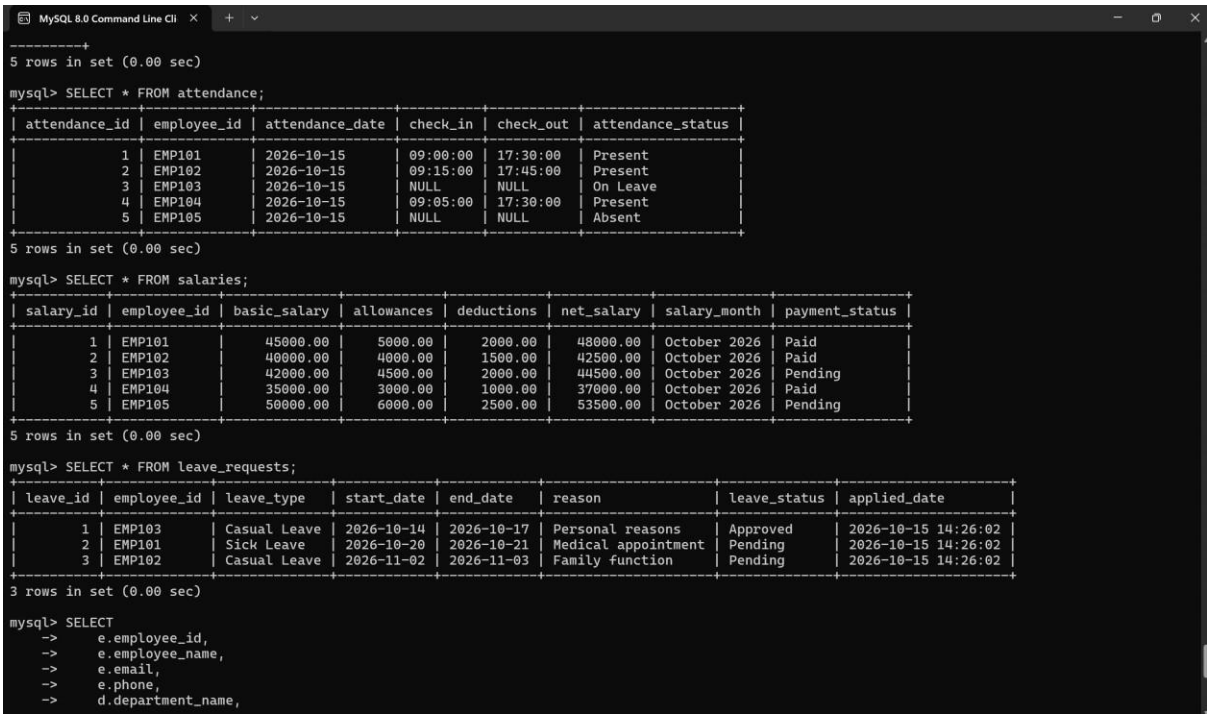
mysql> CREATE TABLE leave_requests (
    ->     leave_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     employee_id VARCHAR(20) NOT NULL,
    ->     leave_type VARCHAR(50),
    ->     start_date DATE NOT NULL,
    ->     end_date DATE NOT NULL,
    ->     reason VARCHAR(255),
    ->     leave_status VARCHAR(30) DEFAULT 'Pending',
    ->     applied_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->
    ->     FOREIGN KEY (employee_id)
    ->     REFERENCES employees(employee_id)
    -> );
Query OK, 0 rows affected (0.12 sec)

mysql> INSERT INTO departments
    -> (department_name, department_description)
    -> VALUES
    -> ('Engineering', 'Software development and maintenance'),
    -> ('Quality Assurance', 'Software testing and quality control'),
    -> ('Design', 'UI and UX designing'),
    -> ('Human Resources', 'Employee management and recruitment'),
    -> ('Finance', 'Financial and salary management'),
    -> ('Marketing', 'Product marketing and promotion'),
    -> ('IT Support', 'Technical support and infrastructure');
Query OK, 7 rows affected (0.10 sec)
Records: 7  Duplicates: 0  Warnings: 0
```


MySQL 8.0 Command Line Cli
mysql> INSERT INTO employees
-> (employee_id, employee_name, email, phone,
-> department_id, job_role, joining_date, employment_status)
-> VALUES
->
-> ('EMP101', 'Aarav Patel', 'aarav@example.com',
-> '9876543210', 1, 'Software Developer', '2023-02-13', 'Active'),
->
-> ('EMP102', 'Priya Sharma', 'priya@example.com',
-> '9876543211', 2, 'QA Engineer', '2022-08-22', 'Active'),
->
-> ('EMP103', 'Rohan Das', 'rohan@example.com',
-> '9876543212', 3, 'UI/UX Designer', '2024-01-08', 'On Leave'),
->
-> ('EMP104', 'Meera Nair', 'meera@example.com',
-> '9876543213', 4, 'HR Executive', '2023-06-19', 'Active'),
->
-> ('EMP105', 'Kabir Singh', 'kabir@example.com',
-> '9876543214', 1, 'Backend Developer', '2021-11-01', 'Inactive');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
mysql> INSERT INTO attendance
-> (employee_id, attendance_date, check_in, check_out, attendance_status)
-> VALUES
->
-> ('EMP101', '2026-10-15', '09:00:00', '17:30:00', 'Present'),
->
-> ('EMP102', '2026-10-15', '09:15:00', '17:45:00', 'Present'),
->
-> ('EMP103', '2026-10-15', NULL, NULL, 'On Leave'),
->
-> ('EMP104', '2026-10-15', '09:05:00', '17:30:00', 'Present'),
->
-> ('EMP105', '2026-10-15', NULL, NULL, 'Absent');
Query OK, 5 rows affected (0.05 sec)
Records: 5 Duplicates: 0 Warnings: 0
mysql> INSERT INTO salaries
-> (employee_id, basic_salary, allowances,
-> deductions, net_salary, salary_month, payment_status)
-> VALUES


MySQL 8.0 Command Line Cli
mysql> INSERT INTO salaries
-> (employee_id, basic_salary, allowances,
-> deductions, net_salary, salary_month, payment_status)
-> VALUES
->
-> ('EMP101', 45000, 5000, 2000, 48000, 'October 2026', 'Paid'),
->
-> ('EMP102', 40000, 4000, 1500, 42500, 'October 2026', 'Paid'),
->
-> ('EMP103', 42000, 4500, 2000, 44500, 'October 2026', 'Pending'),
->
-> ('EMP104', 35000, 3000, 1000, 37000, 'October 2026', 'Paid'),
->
-> ('EMP105', 50000, 6000, 2500, 53500, 'October 2026', 'Pending');
Query OK, 5 rows affected (0.14 sec)
Records: 5 Duplicates: 0 Warnings: 0
mysql> INSERT INTO leave_requests
-> (employee_id, leave_type, start_date,
-> end_date, reason, leave_status)
-> VALUES
->
-> ('EMP103', 'Casual Leave',
-> '2026-10-14', '2026-10-17',
-> 'Personal reasons', 'Approved'),
->
-> ('EMP101', 'Sick Leave',
-> '2026-10-20', '2026-10-21',
-> 'Medical appointment', 'Pending'),
->
-> ('EMP102', 'Casual Leave',
-> '2026-11-02', '2026-11-03',
-> 'Family function', 'Pending');
Query OK, 3 rows affected (0.09 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> SELECT * FROM departments;
+---------------+------------------+-------------------------------------+
| department_id | department_name | department_description |
+---------------+------------------+-------------------------------------+
| 1 | Engineering | Software development and maintenance |
| 2 | Quality Assurance | Software testing and quality control |


MySQL 8.0 Command Line Cli
mysql> SELECT * FROM departments;
+---------------+------------------+-------------------------------------+
| department_id | department_name | department_description |
+---------------+------------------+-------------------------------------+
1	Engineering	Software development and maintenance
2	Quality Assurance	Software testing and quality control
3	Design	UI and UX designing
4	Human Resources	Employee management and recruitment
5	Finance	Financial and salary management
6	Marketing	Product marketing and promotion
7	IT Support	Technical support and infrastructure
+---------------+------------------+-------------------------------------+
7 rows in set (0.00 sec)
mysql> SELECT * FROM employees;
+-------------+---------------+-------------------+------------+---------------+--------------------+--------------+-----------------+---------------------+
| employee_id | employee_name | email | phone | department_id | job_role | joining_date | employment_status | created_at |
+-------------+---------------+-------------------+------------+---------------+--------------------+--------------+-----------------+---------------------+
EMP101	Aarav Patel	aarav@example.com	9876543210	1	Software Developer	2023-02-13	Active	2026-10-15 14:25:32
EMP102	Priya Sharma	priya@example.com	9876543211	2	QA Engineer	2022-08-22	Active	2026-10-15 14:25:32
EMP103	Rohan Das	rohan@example.com	9876543212	3	UI/UX Designer	2024-01-08	On Leave	2026-10-15 14:25:32
EMP104	Meera Nair	meera@example.com	9876543213	4	HR Executive	2023-06-19	Active	2026-10-15 14:25:32
EMP105	Kabir Singh	kabir@example.com	9876543214	1	Backend Developer	2021-11-01	Inactive	2026-10-15 14:25:32
+-------------+---------------+-------------------+------------+---------------+--------------------+--------------+-----------------+---------------------+
5 rows in set (0.00 sec)
mysql> SELECT * FROM attendance;
+---------------+-------------+-----------------+----------+-----------+-------------------+
| attendance_id | employee_id | attendance_date | check_in | check_out | attendance_status |
+---------------+-------------+-----------------+----------+-----------+-------------------+
1	EMP101	2026-10-15	09:00:00	17:30:00	Present
2	EMP102	2026-10-15	09:15:00	17:45:00	Present
3	EMP103	2026-10-15	NULL	NULL	On Leave
Show desktop


MySQL 8.0 Command Line Cli
5 rows in set (0.00 sec)
mysql> SELECT * FROM attendance;
+---------------+-------------+-----------------+----------+-----------+-------------------+
| attendance_id | employee_id | attendance_date | check_in | check_out | attendance_status |
+---------------+-------------+-----------------+----------+-----------+-------------------+
1	EMP101	2026-10-15	09:00:00	17:30:00	Present
2	EMP102	2026-10-15	09:15:00	17:45:00	Present
3	EMP103	2026-10-15	NULL	NULL	On Leave
4	EMP104	2026-10-15	09:05:00	17:30:00	Present
5	EMP105	2026-10-15	NULL	NULL	Absent
+---------------+-------------+-----------------+----------+-----------+-------------------+
5 rows in set (0.00 sec)
mysql> SELECT * FROM salaries;
+-----------+-------------+--------------+------------+------------+------------+--------------+----------------+
| salary_id | employee_id | basic_salary | allowances | deductions | net_salary | salary_month | payment_status |
+-----------+-------------+--------------+------------+------------+------------+--------------+----------------+
1	EMP101	45000.00	5000.00	2000.00	48000.00	October 2026	Paid
2	EMP102	40000.00	4000.00	1500.00	42500.00	October 2026	Paid
3	EMP103	42000.00	4500.00	2000.00	44500.00	October 2026	Pending
4	EMP104	35000.00	3000.00	1000.00	37000.00	October 2026	Paid
5	EMP105	50000.00	6000.00	2500.00	53500.00	October 2026	Pending
+-----------+-------------+--------------+------------+------------+------------+--------------+----------------+
5 rows in set (0.00 sec)
mysql> SELECT * FROM leave_requests;
+----------+-------------+---------------+------------+------------+---------------------+--------------+---------------------+
| leave_id | employee_id | leave_type | start_date | end_date | reason | leave_status | applied_date |
+----------+-------------+---------------+------------+------------+---------------------+--------------+---------------------+
1	EMP103	Casual Leave	2026-10-14	2026-10-17	Personal reasons	Approved	2026-10-15 14:26:02
2	EMP101	Sick Leave	2026-10-20	2026-10-21	Medical appointment	Pending	2026-10-15 14:26:02
3	EMP102	Casual Leave	2026-11-02	2026-11-03	Family function	Pending	2026-10-15 14:26:02
+----------+-------------+---------------+------------+------------+---------------------+--------------+---------------------+
3 rows in set (0.00 sec)
mysql> SELECT
-> e.employee_id,
-> e.employee_name,
-> e.email,
-> e.phone,
-> d.department_name,

```
MySQL 8.0 Command Line Cli

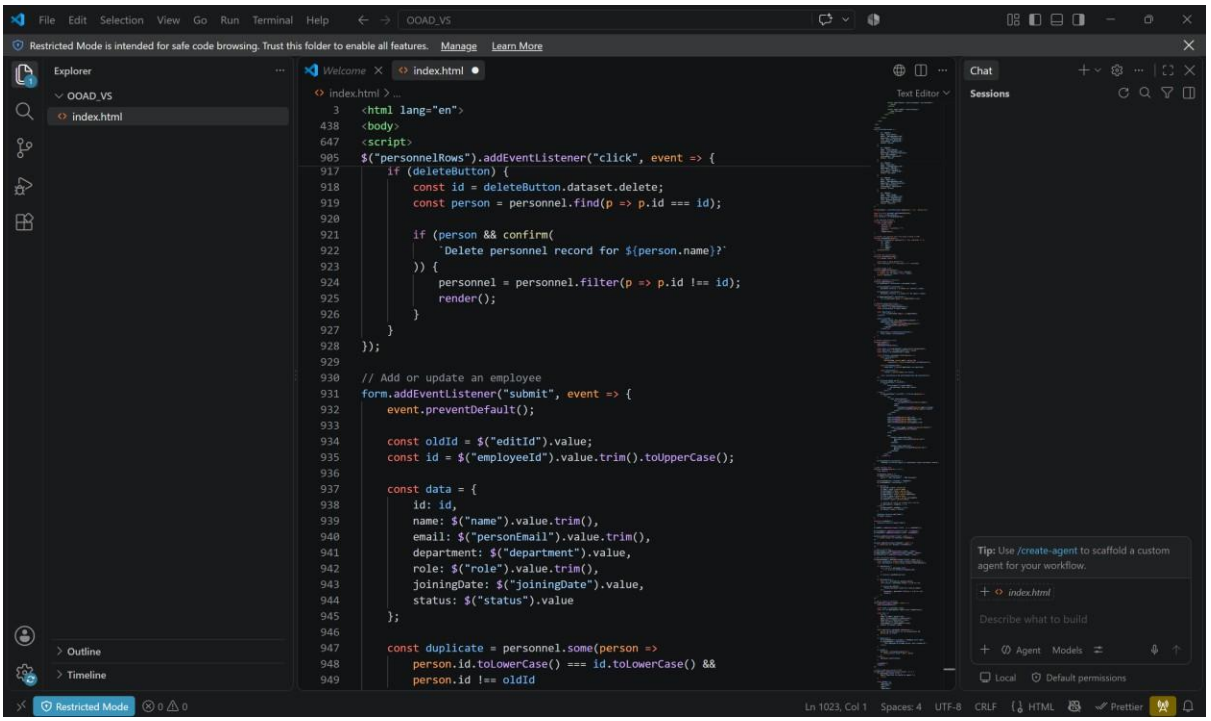
|         4 | EMP104      |     35000.00 |    3000.00 |    1000.00 |   37000.00 | October 2026 | Paid           |
|         5 | EMP105      |     50000.00 |    6000.00 |    2500.00 |   53500.00 | October 2026 | Pending        |
+-----------+-------------+--------------+------------+------------+------------+--------------+----------------+
5 rows in set (0.00 sec)

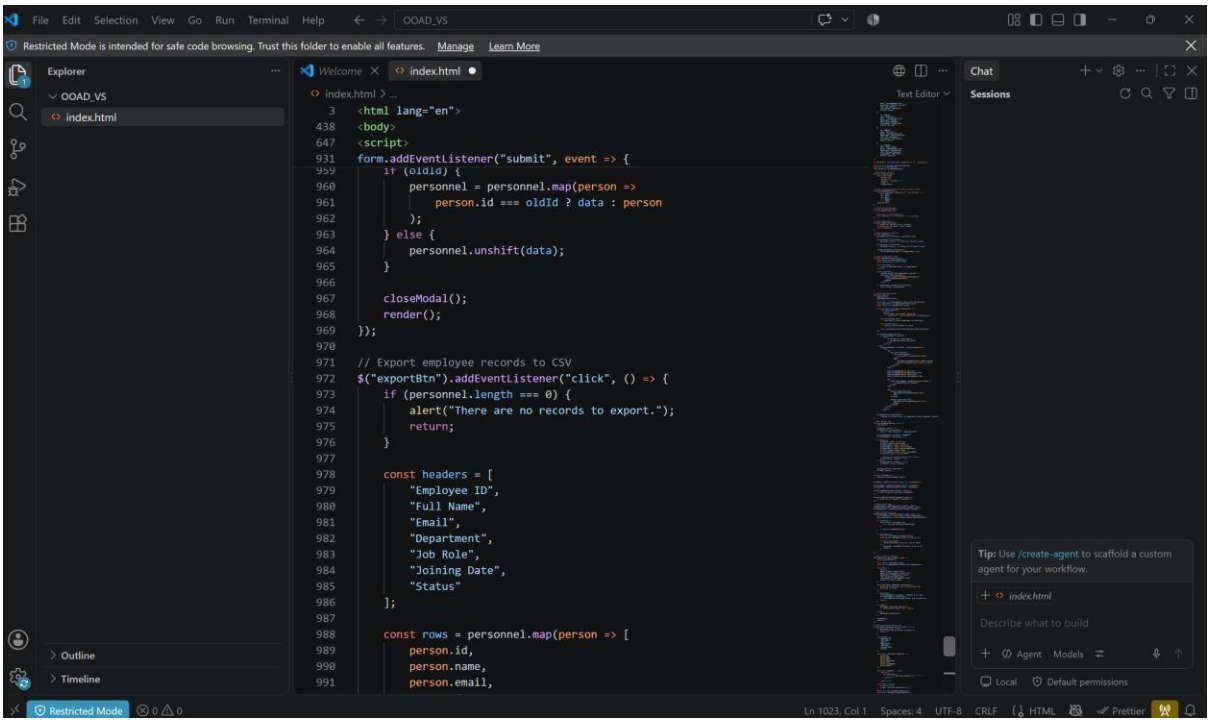
mysql> SELECT * FROM leave_requests;
+----------+-------------+---------------+------------+------------+--------------------+--------------+---------------------+
| leave_id | employee_id | leave_type    | start_date | end_date   | reason             | leave_status | applied_date        |
+----------+-------------+---------------+------------+------------+--------------------+--------------+---------------------+
|        1 | EMP103      | Casual Leave  | 2026-10-14 | 2026-10-17 | Personal reasons   | Approved     | 2026-10-15 14:26:02 |
|        2 | EMP101      | Sick Leave    | 2026-10-20 | 2026-10-21 | Medical appointment | Pending     | 2026-10-15 14:26:02 |
|        3 | EMP102      | Casual Leave  | 2026-11-02 | 2026-11-03 | Family function    | Pending      | 2026-10-15 14:26:02 |
+----------+-------------+---------------+------------+------------+--------------------+--------------+---------------------+
3 rows in set (0.00 sec)

mysql> SELECT
    ->     e.employee_id,
    ->     e.employee_name,
    ->     e.email,
    ->     e.phone,
    ->     d.department_name,
    ->     e.job_role,
    ->     e.joining_date,
    ->     e.employment_status
    ->
    -> FROM employees e
    ->
    -> JOIN departments d
    -> ON e.department_id = d.department_id;
+-------------+---------------+-------------------+------------+-----------------+--------------------+--------------+-------------------+
| employee_id | employee_name | email             | phone      | department_name | job_role           | joining_date | employment_status |
+-------------+---------------+-------------------+------------+-----------------+--------------------+--------------+-------------------+
| EMP101      | Aarav Patel   | aarav@example.com | 9876543210 | Engineering     | Software Developer | 2023-02-13   | Active            |
| EMP102      | Priya Sharma  | priya@example.com | 9876543211 | Quality Assurance | QA Engineer      | 2022-08-22   | Active            |
| EMP103      | Rohan Das     | rohan@example.com | 9876543212 | Design          | UI/UX Designer     | 2024-01-08   | On Leave          |
| EMP104      | Meera Nair    | meera@example.com | 9876543213 | Human Resources | HR Executive       | 2023-06-19   | Active            |
| EMP105      | Kabir Singh   | kabir@example.com | 9876543214 | Engineering     | Backend Developer  | 2021-11-01   | Inactive          |
+-------------+---------------+-------------------+------------+-----------------+--------------------+--------------+-------------------+
5 rows in set (0.00 sec)

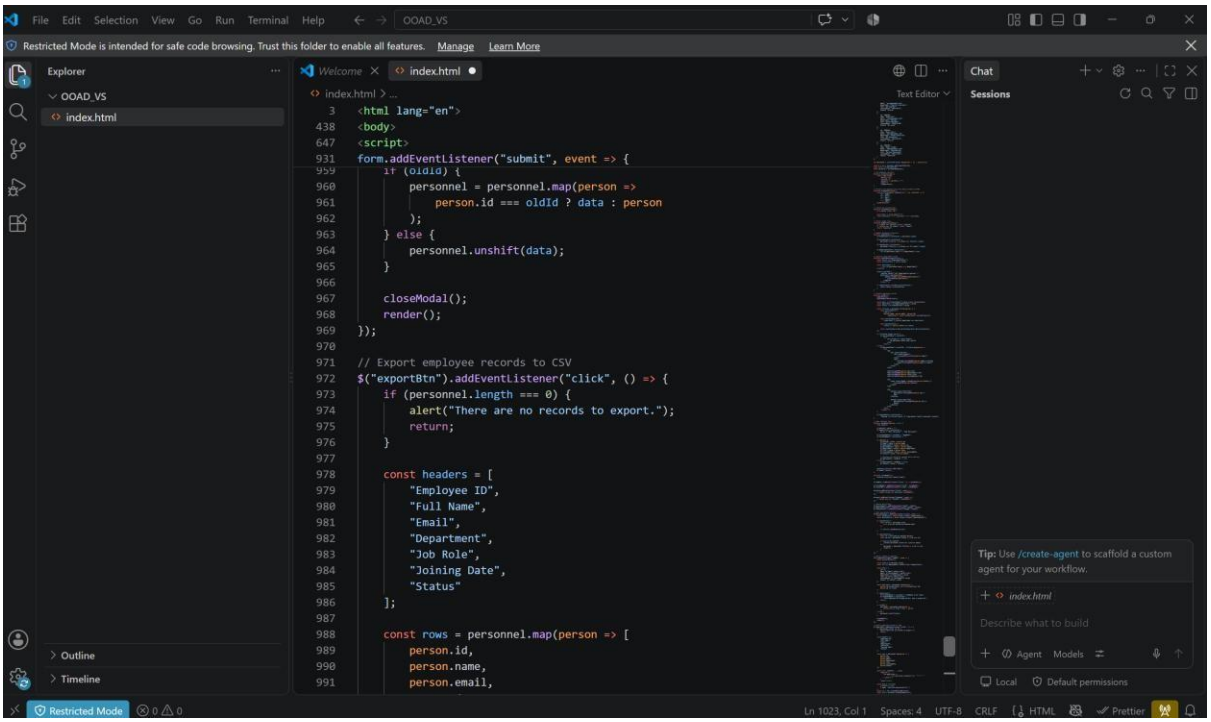
mysql>
```

HTML CODE:






```
3   <html lang="en">
438 <body>
647 <script>
931 form.addEventListener("submit", event => {
959     if (oldId) {
960         personnel = personnel.map(person =>
961             person.id === oldId ? data : person
962         );
963     } else {
964         personnel.unshift(data);
965     }
966
967     closeModal();
968     render();
969 });
970
971 // Export employee records to CSV
972 $("exportBtn").addEventListener("click", () => {
973     if (personnel.length === 0) {
974         alert("There are no records to export.");
975         return;
976     }
977
978     const headers = [
979         "Employee ID",
980         "Full Name",
981         "Email",
982         "Department",
983         "Job Role",
984         "Joining Date",
985         "Status"
986     ];
987
988     const rows = personnel.map(person => [
989         person.id,
990         person.name,
991         person.email,
```




```
3   <html lang="en">
438 <body>
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